THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of angles around a point is 360°. If three angles are 95°, 85°, 110° the fourth is: 1

(a) 80° **(b)** 90° **(c)** 70° **(d)** 60°

2. If x and 68° are complementary, $x =$ 1

(a) 22° **(b)** 42° **(c)** 32° **(d)** 12°

3. In a linear pair, one angle is 8 times the other. The smaller angle is: 1

(a) 10° **(b)** 40° **(c)** 30° **(d)** 20°

4. Two parallel lines are cut by a transversal. If an interior angle is 135°, the co-interior angle on the same side is: 1

(a) 65° **(b)** 45° **(c)** 55° **(d)** 35°

5. The number of angles formed by a transversal on two lines is: 1

(a) 6 **(b)** 12 **(c)** 8 **(d)** 4

6. Angle $A = 143^\circ$ and angle B are vertically opposite. Angle B is: 1

(a) 143° **(b)** 163° **(c)** 153° **(d)** 133°

7. An angle of 210° is: 1

(a) acute **(b)** right **(c)** reflex **(d)** obtuse

8. The sum of angles around a point is 360°. If three angles are 130°, 65°, 75° the fourth is: 1

(a) 80° **(b)** 100° **(c)** 90° **(d)** 110°

9. If x and 83° are complementary, $x =$ 1

(a) 7° **(b)** 17° **(c)** 37° **(d)** 27°

10. An angle of 89° is: 1

(a) right **(b)** acute **(c)** obtuse **(d)** reflex

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two angles are supplementary and one is 4 times the other. Find them. 2

12. Find x if $(2x + 6)^\circ$ and (94°) form a linear pair. 2

13. Lines $l \parallel m$ and a transversal. A pair of alternate interior angles are $(x + 30)^\circ$ and 95°. Find x. 2

14. Two angles are supplementary and one is 5 times the other. Find them. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Lines $l \parallel m$, transversal t. Angle $\angle 1 = (3x + 20)^\circ$ and its corresponding angle $= (2x + 70)^\circ$. Find x and the angle. 3

16. Co-interior angles on parallel lines are $(4x + 8)^\circ$ and $(2x + 10)^\circ$. Find x and both angles. 3

17. Two lines intersect forming angles $(5x - 4)^\circ$ and $(3x + 16)^\circ$ as vertically opposite angles. Find x and all four angles. 3

OR

Lines $l \parallel m$, transversal t. Angle $\angle 1 = (3x + 12)^\circ$ and its corresponding angle $= (2x + 50)^\circ$. Find x and the angle. 3

18. Co-interior angles on parallel lines are $(4x + 6)^\circ$ and $(2x + 24)^\circ$. Find x and both angles. 3

OR

Two lines intersect forming angles $(5x - 2)^\circ$ and $(3x + 12)^\circ$ as vertically opposite angles. Find x and all four angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Lines $l \parallel m$ and transversal t. Angles at l: $\angle 1 = 125^\circ$. Write the values of the corresponding, alternate interior, vertically opposite and co-interior angles of $\angle 1$ at m, with reasons. 5

OR

Two parallel lines are cut by a transversal and one of the eight angles is 125°. How many of the eight angles measure 125° and how many measure 55°? Give reasons. 5

20. Lines $l \parallel m$ and transversal t. Angles at l: $\angle 1 = 140^\circ$. Write the values of the corresponding, alternate interior, vertically opposite and co-interior angles of $\angle 1$ at m, with reasons. 5

OR

Two parallel lines are cut by a transversal and one of the eight angles is 140°. How many of the eight angles measure 140° and how many measure 40°? Give reasons. 5

END OF QUESTION PAPER — Total: 40 Marks

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